# Kishoreganj University

**2nd Year 2nd Semester B. Sc. (Engg.) Final Examination-2023**

**Department of Computer Science and Engineering**

**CSE 2203: Numerical Methods (3 Credits)**

**Time: 3 Hours** **Full Marks: 70**

***Figures shown in the right margin indicate full marks.***
***Answer 05 out of 07 questions.***

**1.** **a.** Describe the concept applied in the bracketing methods used for solving nonlinear equations. Use the False Position Method to obtain a root of the following equation by performing five iterations. **6**

$$3x^2 + 6x - 45 = 0$$

**b.** Obtain the quadratic factor of the following polynomial using Bairstow's method with starting values $u = +1.8$ and $v = -4$. **4**

$$p(x) = x^3 + x + 10$$

**c.** Compute a root of the following equation using Newton Raphson Method. Perform five iterations. **4**

$$x^4 + 3x^3 - 2x^2 - 12x - 8 = 0,\ x_0 = 1$$

**2.** **a.** A system of resistors in an electrical circuit follows Kirchhoff's Current Law (KCL), leading to the following system of linear equations for the currents $I_1, I_2, I_3$: **6**

$$2I_1 + I_2 - I_3 = 3$$
$$I_1 + 3I_2 + 2I_3 = 9$$
$$3I_1 + 2I_2 + 4I_3 = 13$$

Identify the value of current flows $I_1, I_2, I_3$ using the Gauss-Jordan method.

**b.** Explain Crout's Decomposition Method. Decompose the following matrix using the Crout's method. **4**

$$A=\begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix}$$

**c.** Decompose the matrix **4**

$$A = \begin{bmatrix} 4 & 3 & -1 \\ 1 & 1 & 1 \\ 3 & 5 & 3 \end{bmatrix}$$

into the form LU where L is unit lower triangular and U is upper triangular matrices.

**3.** **a.** Using Taylor's Expansion, derive a formula for computing the second derivative of a function. Find approximation to second derivative of the following function: **6**

$$f(x) = e^x \sin(x)$$

at $x = 0.75$ with $h = 0.01$. Compare the result with the true value.

**b.** Compute the approximate derivatives of $f(x) = sinx + cosx$ at $x = 0.45$ radians using the first order forward difference formula, at increasing values of $h$ from $0.01$ to $0.04$, with a step-size of $0.005$. Analyze the total error. Identify the step where the error is minimum. **4**

**c.** Explain Richardson's Extrapolation. "Richardson's Extrapolation improves the estimates of derivatives"- justify the statement. **4**

**4.** **a.** Summarize the necessity of pivoting for solution of linear systems. **3**

**b.** State the differences between Simpson's 1/3 rule and Simpson's 3/8 rule for numerical integration. Compute the values of **7**

$$I = \int_0^1 \frac{dx}{1+x^2}$$

using both Trapezoidal and Simpson's ⅓ rule with h=0.25. Also compare between the obtained results.

c. Explain Newton Cote's Method for integration with proper examples. How could we improve the accuracy of a numerical integration process? 4

5. a. Explain the term 'interpolation' with proper example. 3

b. The population of Mississippi during three census periods was as follows: 5

| Year | 1951 | 1961 | 1971 |
|---|---|---|---|
| Population (Million) | 2.8 | 3.2 | 4.5 |

Estimate the population during 1966 using Lagrange's interpolation formula.

c. Identify the cubic polynomial using both forward difference and backward difference formula which takes the following values $y(1) = 24$, $y(3) = 120$, $y(5) = 336$ and $y(7) = 720$. Also, compare the results obtaining from both formulas. 6

6. a. In an organization, systematic efforts were introduced to reduce the employee absenteeism and results for the first 10 months are shown below: 6

| Months | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Absentees (%) | 10 | 9 | 9 | 8.5 | 9 | 8 | 8.5 | 7 | 8 | 7.5 |

Fit a linear least squares line to the data and plot the data and the regression line in a graph.

b. Fit the power equation $y = ax^b$ to the following data: 4

| x | 2 | 4 | 6 | 8 |
|---|---|---|---|---|
| y | 1.4 | 2.0 | 2.4 | 2.6 |

c. Fit a second order polynomial to the data in the table below: 4

| x | 1.0 | 2.0 | 3.0 | 4.0 |
|---|---|---|---|---|
| y | 6.0 | 11.0 | 18.0 | 27.0 |

7. a. Use the simple Euler's method to solve the following equations for $y(1)$ using $h = 0.5$. 6

(i) $\frac{dy}{dx} = x^2 + y^2$, where $y(0) = 2$

(ii) $\frac{dy}{dx} = x + y + xy$, where $y(0) = 1$

b. Find $y(0.5)$ for $\frac{dy}{dx} = -4x - y$, where $y(0) = -1$, with step length 0.1 using Taylor's expansion method. 4

c. The general equation relating to current $i$, voltage $V$, resistance $R$, and inductance $L$ of a serial electric circuit is given by 4

$$L\frac{di}{dt} + iR = V$$

Find the value of current after 2 seconds, if resistance $R = 20\Omega$, inductance $L = 50H$, and voltage $V = 240V$. Consider that current $I = 0$ when $t = 0$.